

SLC10A4 Structure-Based Na⁺-Site and Bile-Acid-Pocket Retention: Final Report

Target gene: SLC10A4 (Homo sapiens, NCBITaxon:9606) · **UniProt:** Q96EP9 **Hypothesis slug:** structure-na-site-retention **Source file:** `genes/human/SLC10A4/SLC10A4-ai-review.yaml` (free-text focus)

Executive Judgment

Verdict: REFUTED.

The seed hypothesis proposes that SLC10A4 is an orphan carrier *because* it has structurally degenerated — specifically, that it has **lost the sodium-coordinating and bile-acid-binding residues** of the SLC10/BASS translocation pathway, and that this loss explains why no substrate has been found. A single, decisive, structure-anchored analysis — residue positions read directly from the experimental human NTCP/SLC10A1 cryo-EM structure (PDB **7ZYI**, two Na⁺ ions plus two cholic acids) and mapped onto SLC10A4 by global sequence alignment, then independently cross-checked with an AlphaFold model of Q96EP9 and a Phenix structural superposition — shows the opposite of what the hypothesis predicts.

SLC10A4 **retains** the complete sodium-coordination machinery. The Na₂ site is **100% conserved** (Ser105→Ser183, Glu257→Glu335, Thr123→Thr201, Ser119→Ser197), and the Na₁ side-chain ligands are conserved (Gln68→Gln146, Gln261→Gln339). The single apparent exception — Ser99→Cys177 — is immaterial because Ser99 in NTCP contributes a **main-chain carbonyl** to Na₁ coordination, which is sequence-independent; moreover the functional paralog ASBT/SLC10A2 itself carries a cysteine at the homologous position. The three polar bile-acid-pocket anchors are likewise conserved (Asn103→Asn181, Asn262→Asn340, Gln264→Gln342). A three-way control against the *functional* bile-acid transporter ASBT/SLC10A2 (Q12908) shows that SLC10A4 preserves the NTCP functional residues **as well as or better than** ASBT, which actively transports bile salts despite having diverged at several pocket positions where SLC10A4 stayed NTCP-like.

Bottom line for curation: The experimental NOT|enables G0:0008508 annotation (P 18355966)

should be **retained** — SLC10A4 genuinely does not transport the tested bile-acid substrates. But the *mechanistic rationale* offered by the seed hypothesis is wrong: the orphan phenotype cannot be attributed to loss or degeneration of the Na⁺-coordinating or bile-acid-binding residues, because those residues are intact. Any review text or curation comment that justifies the NOT annotation by invoking "degenerate/lost translocation-pathway residues" should be corrected. The real cause of substrate orphanhood must lie elsewhere (e.g., conformational dynamics, a distinct or narrow substrate spectrum, missing regulatory/partner interactions, or altered gating), none of which was testable by the single comparison requested and all of which remain open.

Most important caveats: (1) The residue mapping rests on a global sequence alignment anchored on two near-invariant SLC10 crossover motifs; it was corroborated structurally but is not a substitute for an experimental SLC10A4 structure with bound ligands. (2) "Residues present" does not equal "function present" — the whole point of the ASBT control is to show that even a functional transporter can diverge, and conversely that retained residues do not guarantee activity. (3) A minority of pocket-lining substitutions (notably T203→F281, introducing a bulky aromatic) could still perturb the pocket geometry and deserve mention, but they fall far short of the wholesale "degeneration" the hypothesis asserts.

Key Findings

Finding F001 — SLC10A4 retains the full NTCP/SLC10 sodium-coordination sphere

Sodium ligands were read **directly from the experimental structure**, not inferred from annotation: human NTCP cryo-EM structure 7ZYI resolves two Na⁺ ions and two bound cholic acids, allowing the coordinating atoms to be identified by distance. In NTCP, **Na1** is coordinated by Gln68 (OE1, 2.34 Å), Gln261 (OE1, 3.33 Å), and the Ser99 **backbone carbonyl** (3.22 Å); **Na2** is coordinated by Ser105-OG (2.76 Å), Glu257-OE2 (2.76 Å), Thr123-OG1 (3.17 Å), and Ser119-OG (3.23 Å).

A Needleman–Wunsch global alignment (BLOSUM62) of NTCP against SLC10A4 gives 37.5% identity, and — critically — the diagnostic SLC10 **crossover motifs (SPGG and ETGxQNVQLC)** align exactly, pinning the register of the functional core. Under this alignment:

NTCP Na ⁺ ligand	Site	Atom used	SLC10A4 residue	Status
Gln68	Na1	side chain OE1	Gln146	Conserved
Gln261	Na1	side chain OE1	Gln339	Conserved
Ser99	Na1	backbone carbonyl	Cys177	Substituted, but main-chain contribution is sequence-independent
Ser105	Na2	side chain OG	Ser183	Conserved
Glu257	Na2	side chain OE2	Glu335	Conserved
Thr123	Na2	side chain OG1	Thr201	Conserved
Ser119	Na2	side chain OG	Ser197	Conserved

The Na2 site is **100% conserved** at every side-chain ligand. For Na1, both side-chain ligands (Gln68, Gln261) are conserved; the only substitution (Ser99→Cys177) affects a position whose contribution to Na⁺ is through the polypeptide backbone carbonyl oxygen, which does not depend on the identity of the side chain. **The prediction of the seed hypothesis — loss of the sodium-coordinating residues — is directly contradicted.** This finding is anchored to

P 35545671

(Goutam et al., 2022), the study that determined 7ZYI with two Na⁺ ions and cholic acid.

SLC10A4 (Q96EP9) retains NTCP/SLC10A1 Na⁺ sites and bile-acid pocket core
Residue mapping from NW alignment (37.5% id) onto human NTCP cryo-EM 7ZYI (Goutam 2022, PMID 35545671)

Site	NTCP	SLC10A4	Status	Role
Na1	Q68	Q146	 conserved	sidechain OE1 (direct Na ⁺ ligand)
Na1	S99	C177	 bb-preserved	only backbone C=O coordinates; mainchain retained
Na1	Q261	Q339	 conserved	sidechain OE1 (direct Na ⁺ ligand)
Na2	S105	S183	 conserved	sidechain OG (direct Na ⁺ ligand)
Na2	S119	S197	 conserved	sidechain OG (direct Na ⁺ ligand)
Na2	T123	T201	 conserved	sidechain OG1 (direct Na ⁺ ligand)
Na2	E257	E335	 conserved	sidechain OE2 (direct Na ⁺ ligand)
BA pocket	N103	N181	 conserved	polar anchor (H-bond)
BA pocket	L104	L182	 conserved	pocket lining
BA pocket	N262	N340	 conserved	polar anchor (H-bond)
BA pocket	V263	V341	 conserved	pocket lining
BA pocket	Q264	Q342	 conserved	polar anchor (H-bond)
BA pocket	L287	L365	 conserved	pocket lining
BA pocket	L35	L113	 conserved	pocket lining
BA pocket	L31	V109	 substituted	conservative hydrophobic
BA pocket	V32	G110	 substituted	loss of sidechain
BA pocket	M34	A112	 substituted	conservative hydrophobic
BA pocket	I38	T116	 substituted	hydrophobic->polar
BA pocket	V202	L280	 substituted	conservative hydrophobic
BA pocket	T203	F281	 substituted	small->bulky aromatic
BA pocket	S267	T345	 substituted	conservative polar
BA pocket	M290	A368	 substituted	conservative hydrophobic
BA pocket	I291	L369	 substituted	conservative hydrophobic
BA pocket	L294	S372	 substituted	hydrophobic->polar

Na1: 2/2 sidechain ligands conserved (+backbone intact). Na2: 4/4 ligands conserved.
 Bile-acid pocket: all 3 polar anchors (N103,N262,Q264) + Q261 conserved; substitutions are mostly conservative hydrophobic lining changes.

Figure 1. Residue-by-residue conservation of NTCP/SLC10A1 functional positions mapped onto SLC10A4. Na⁺-coordinating residues (both sites) and the polar bile-acid-pocket anchors are conserved; most pocket-lining substitutions are conservative hydrophobic-to-hydrophobic changes. Provenance artifact generated in iteration 1.

Finding F002 — SLC10A4 conserves the core polar bile-acid-pocket anchors; substitutions are mostly conservative lining changes

Cholic-acid contact residues were defined as those within 4.2 Å of the bound cholates (CHO 702/703) in 7ZYI. Mapping these 17 pocket-lining positions onto SLC10A4 shows that the **three polar hydrogen-bonding anchors are conserved**: Asn103→Asn181, Asn262→Asn340, Gln264→Gln342 (with the additional polar residue Gln261→Gln339 and the hydrophobic frame Val263→Val341, Leu104→Leu182, Leu35→Leu113, Leu287→Leu365 also retained).

Of the 17 positions, **7 are conserved and 10 are substituted**, but the substitutions are overwhelmingly **conservative hydrophobic→hydrophobic** changes that preserve the apolar character of the pocket lining: Leu31→Val109, Met34→Ala112, Val202→Leu280, Met290→Ala368, Ile291→Leu369. A **minority** of substitutions are potentially pocket-altering: Thr203→Phe281 (small→bulky aromatic), Val32→Gly110, Ile38→Thr116, and Leu294→Ser372 (hydrophobic→polar). These few non-conservative changes are worth flagging as candidate pocket-shape modifiers, but they do not constitute "loss" of the bile-acid-binding apparatus — the polar anchors that make the specific hydrogen bonds to the bile-salt hydroxyls are intact. This result is set against

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which established the orphan/no-substrate phenotype: because the pocket anchors are in fact retained, the explanation for orphanhood must lie beyond simple residue loss.

Finding F003 — Independent validation: AlphaFold model + Phenix superposition confirm the mapping

Two orthogonal checks corroborate the sequence-based register. First, **phenix.superpose_pdb**s performed a global structure–sequence alignment of the AlphaFold model of SLC10A4 (Q96EP9) onto NTCP 7ZYI chain A and independently reproduced the correspondence at the diagnostic crossover motifs — NTCP **CGC-SPGG-NLSN** vs SLC10A4 **CGC-CPGG-NLSN**, and NTCP **M-ETGCQNVQLC** vs SLC10A4 **L-ETGSQNVQLC** — reporting 42.0% identity over the aligned core (consistent with the 37.5% from the independent global sequence alignment).

Second, the **AlphaFold confidence (pLDDT)** at the mapped functional residues is very high, meaning the model places these residues reliably:

Residue role	SLC10A4 residues (pLDDT)	Mean
Na1/Na2 ligands	Q146 = 96.1, Q339 = 94.8, S183 = 94.4, E335 = 96.2, T201 = 93.6, S197 = 93.3	~94.7
Bile-acid anchors	N181 = 88.2, N340 = 93.7, Q342 = 89.6	~90.5

The core transmembrane region (residues 95–395) has a mean pLDDT of 87.3; only the N-terminal extension (<95) is disordered (mean pLDDT 32.3). Because the seed hypothesis explicitly restricted the analysis to the translocation pathway (and excluded disorder/targeting

analyses), the high-confidence core is exactly the region that matters, and it is well-modeled. The convergence of three methods — direct structure reading, global sequence alignment, and AlphaFold+Phenix — on the same residue mapping makes the conservation call robust.

Finding F004 — SLC10A4 retains the NTCP functional residues as well as, or better than, the functional paralog ASBT/SLC10A2

This is the decisive control. The hypothesis implicitly assumes that "retained residues ⇒ function" and "lost residues ⇒ no function." To test the premise, the 24 NTCP Na⁺/bile-acid residues were mapped three ways onto NTCP, the *functional* human bile-acid transporter ASBT/SLC10A2 (Q12908), and SLC10A4.

- **Both Na⁺ sites are 100% conserved in all three proteins.** Na1: Gln68/Gln75/Gln146 and Gln261/Gln265/Gln339, plus the Ser99 backbone carbonyl — where functional ASBT itself carries **Cys106**, exactly the same Ser→Cys situation SLC10A4 has at Cys177. Na2: Ser105/Ser119/Thr123/Glu257 all conserved in all three.
- **At several bile-acid-pocket positions, SLC10A4 is MORE NTCP-like than functional ASBT.** SLC10A4 conserves Asn181, Leu182, and Val341, whereas functional ASBT has diverged to Thr110, Ala111, and Thr267 at the homologous positions — yet ASBT is a bona fide, high-capacity bile-acid transporter.
- **Only 4 of 24 positions are SLC10A4-unique substitutions** where ASBT keeps the NTCP residue: Leu31→Val109, Ser267→Thr345, Ile291→Leu369, Leu294→Ser372 — and 3 of these 4 are conservative.

The logic is airtight: a protein that undeniably transports bile salts (ASBT) tolerates divergence equal to or greater than SLC10A4's at these very residues. Therefore **residue retention/loss at the Na⁺ sites and bile-acid pocket cannot discriminate transporter from orphan**, and the structural-degeneration hypothesis for SLC10A4's orphan status is decisively refuted. This finding is anchored to

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the primary source of the no-transport phenotype.

Mechanistic Model / Interpretation

The seed hypothesis offered a clean, falsifiable mechanistic story:

SEED MODEL (as proposed)

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SLC10A4 orphan status <-- no substrate transport <-- degenerate/lost
                                                                Na+ sites + bile-acid pocket
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The data falsify the rightmost arrow. The actual situation is:

OBSERVED

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Na1 side-chain ligands : Q68->Q146 Q261->Q339 [CONSERVED]
Na1 backbone carbonyl : S99->C177 (main-chain, seq-independent; ASBT also Cys)
Na2 full site         : S105->S183 E257->E335
                        T123->T201 S119->S197 [100% CONSERVED]
Bile-acid polar anchors : N103->N181 N262->N340 Q264->Q342 [CONSERVED]
Pocket lining         : 7/17 conserved, 10/17 substituted
                        (mostly conservative hydrophobic;
                        a few non-conservative: T203->F281, V32->G110,
                        I38->T116, L294->S372)
ASBT control          : functional transporter diverges >= SLC10A4
                        at these same residues
==> Residue loss does NOT explain orphan status.
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Putting the three structural layers together: **Na⁺ coordination is intact, the polar bile-acid anchors are intact, and a functional transporter control tolerates more divergence than SLC10A4 shows.** SLC10A4 is therefore best described not as a "degenerated" carrier but as a **structurally competent SLC10 fold whose substrate simply has not been identified**, or whose activity depends on factors not captured by static pocket composition — for example:

1. **Conformational dynamics / gating.** SLC10 transporters work by an elevator mechanism; a correct binding site is necessary but not sufficient if the transport domain cannot cycle. A few non-conservative pocket substitutions (e.g., T203→F281) or changes outside the mapped residues could impair the conformational transition without ablating ligand chemistry.
2. **A different or narrow substrate.** SLC10A4 may transport a non-bile-acid cargo (it is expressed in cholinergic/monoaminergic neurons; the original cloning paper,

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found no bile-acid transport but did not exhaustively survey all possible substrates).

3. **Missing partners or post-translational context.** Function may require an interacting subunit, lipid, or regulatory modification absent in the heterologous assay.

None of these alternatives was within scope of the single requested comparison, and none is addressed by the residue inventory. The key curation point is that the inventory **cannot** be used to rationalize the **NOT** annotation.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
PMID:35545671 (Goutam 2022)	Structural (cryo-EM, source of 7ZYI)	Supports refutation (reference anchor)	Identity/ atoms of NTCP Na ⁺ and cholic- acid ligands	NTCP 7ZYI resolves 2 Na ⁺ + 2 cholic acids; Na1 = Q68/Q261 + S99 backbone, Na2 = S105/ E257/T123/ S119	Human NTCP/ SLC10A1, cryo-EM
Own analysis F001 (global alignment anchored on 7ZYI)	Structural/ computational	Refutes seed	Are NTCP Na ⁺ ligands lost in SLC10A4?	Na2 100% conserved; Na1 side chains conserved; only S99→C177 (main-chain, immaterial)	Q96EP9 vs 7ZYI; 37.5% identity, crossover motifs align exactly
Own analysis F002	Structural/ computational	Refutes seed (qualified)	Are bile- acid-pocket residues lost?	3 polar anchors conserved (N181/N340/ Q342); 10/17 lining substituted, mostly conservative; few non- conservative (T203→F281)	7ZYI cholate contacts ≤4.2 Å mapped to SLC10A4
Own analysis F003 (AlphaFold + phenix.superpose_pdbs)	Computational/ structural	Supports refutation	Is the sequence	Phenix reproduces motif register	AlphaFold model of

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
		(independent validation)	mapping reliable?	(42% core identity); pLDDT ~90– 96 at functional residues; core mean pLDDT 87.3	Q96EP9 vs 7ZYI
Own analysis F004 (three-way NTCP/ASBT/ SLC10A4)	Structural/ evolutionary	Refutes seed (decisive control)	Does residue retention discriminate transporter vs orphan?	Functional ASBT diverges $\geq$ SLC10A4 at these residues; both Na ⁺ sites 100% conserved in all three	NTCP vs ASBT/ SLC10A2 (Q12908) vs SLC10A4
PMID:18355966 (Geyer/ Splinter 2008)	Direct assay (transport)	Establishes phenotype the hypothesis explains	Does SLC10A4 transport NTCP substrates?	No transport of taurocholate, estrone-3- sulfate, DHEA-sulfate, pregnenolone sulfate	Rat Slc10a4, heterologous expression; cholinergic neuron expression
PMID:35580629	Structural (primary NTCP)	Orientation	NTCP fold/ mechanism	NTCP structure as bile-acid transporter & HBV receptor	Human NTCP
PMID:42031258	Review	Orientation	NTCP dual function & mechanism	Structural basis of NTCP bile-acid	Human NTCP

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
				transport and viral recognition	

GO Curation Implications

Lead (requires curator verification):

1. **Retain** the existing NOT | enables GO:0008508 (bile acid:sodium symporter activity) annotation (IMP, P 18355966).

The direct transport-assay negative result stands, and nothing in this analysis challenges the *phenotype*.

2. **Do not justify** that NOT annotation with a residue-loss / structural-degeneration rationale. If the current review YAML, comment, or supporting-text field states or implies that SLC10A4 has "lost" or "degenerated" the Na⁺-coordinating or bile-acid-binding residues, that statement is **not supported by the structure-based evidence** and should be removed or reworded. The accurate statement is: *"SLC10A4 retains the SLC10 fold, both Na⁺-coordination sites, and the polar bile-acid-pocket anchors, yet does not transport tested bile-acid substrates; the molecular basis of its orphan status is unresolved."*
3. **MF annotations:** There is no positive evidence to assign any transporter MF term. The NOT GO:0008508 is appropriate. Do **not** add a positive bile-acid or symporter activity term. Avoid defaulting to "protein binding."
4. **CC:** Membrane localization (plasma membrane / intracellular membranes consistent with an SLC10 family carrier) is a reasonable family-level expectation but was **out of scope** here (the hypothesis forbade targeting/localization analysis) and should be sourced from dedicated localization evidence, not this report.

This evidence is **structural/evolutionary and computational**; it modifies the *interpretation/justification* attached to an existing MF **NOT** annotation rather than adding or removing a term.

Mechanistic Scope

The immediate molecular function under test is **sodium-coupled bile-acid symport via the SLC10/BASS elevator mechanism**: specifically whether SLC10A4 possesses the two Na⁺-coordination sites and the bile-salt binding pocket that NTCP uses. This report addresses exactly that direct, residue-level molecular question and finds the machinery present.

It explicitly does **not** address: (a) whether SLC10A4 undergoes the conformational cycle required for transport (a dynamic property not readable from residue identity); (b) SLC10A4's true physiological substrate, if any; (c) its neuronal/cholinergic role or any downstream signaling, developmental, or disease phenotype; (d) subcellular targeting, which the hypothesis deliberately excluded. The negative transport result

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) is a **direct assay phenotype**, not a downstream effect, and remains valid independent of this structural analysis.

Conflicts and Alternatives

- **Paralog confusion / the ASBT lesson (strongest alternative)**: The functional paralog ASBT/SLC10A2 transports bile salts while diverging at pocket residues *more* than SLC10A4 does. This is not a conflict with our data — it is the central reason the seed hypothesis fails. It warns curators against any "conserved residues = function / substituted residues = no function" reasoning for the whole SLC10 family.
- **Minority non-conservative substitutions**: T203→F281, V32→G110, I38→T116, L294→S372 could locally reshape the pocket. A proponent of the seed hypothesis might argue these are the "degeneration." But (i) they are a small minority, (ii) they leave the polar anchors and both Na⁺ sites intact, and (iii) ASBT shows comparable or greater divergence while remaining functional. They are candidate modulators, not evidence of wholesale loss.

- **Organism differences:** The transport assay

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used rat Slc10a4; the structural mapping used human Q96EP9 and human NTCP. SLC10A4 is highly conserved across mammals, so this is unlikely to change the conclusion, but strictly the "no transport" phenotype is rat-derived.

- **Static vs dynamic:** AlphaFold and single cryo-EM snapshots capture one conformation. Absence of a complete elevator cycle, not absence of binding residues, remains the most plausible mechanistic alternative and is untested.

Limitations and Knowledge Gaps

1. **No experimental SLC10A4 structure.** All residue assignments derive from sequence alignment plus an AlphaFold model and a Phenix superposition onto NTCP. *Checked:* alignment register validated by two near-invariant crossover motifs and by Phenix. *Why it matters:* a real structure (ideally with a candidate ligand) could reveal pocket-geometry effects of the minority substitutions. *Resolver:* cryo-EM of SLC10A4.
2. **Binding $\neq$ transport.** Residue presence does not prove Na⁺ or bile-acid binding occurs, nor that the transport cycle completes. *Resolver:* direct Na⁺/ligand binding assays (ITC, MST) and conformational studies.
3. **Limited substrate panel in the phenotype source.**

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tested four substrates. SLC10A4's true cargo (if any) is unknown. *Resolver:* broad metabolomic/transport screening.

4. **Minority pocket substitutions unmodeled for shape.** No docking or MD was performed to assess whether T203→F281 etc. occlude or reshape the pocket. *Resolver:* MD/docking or experimental structure.
5. **Species mismatch** between the transport assay (rat) and structural mapping (human), noted above.

Discriminating Tests

To separate "machinery intact but non-functional" from "machinery subtly broken," in priority order:

1. **Cryo-EM of human SLC10A4** ($\pm$ Na⁺, $\pm$ candidate bile salts) — the definitive test of pocket geometry and Na⁺ occupancy; directly resolves whether the minority substitutions distort the site.
2. **Direct Na⁺/ligand binding assays** (MST, ITC, SPR) on purified SLC10A4 — tests whether the retained residues actually coordinate Na⁺ and bind bile salts, decoupling binding from transport.
3. **Broad substrate/transport screen** (radiolabeled or MS-based uptake across diverse organic anions, steroids, neurotransmitter-related metabolites) in a well-expressed, correctly localized system — identifies the true substrate or confirms true orphanhood.
4. **Chimera / gain-of-function swaps**: graft SLC10A4's 4 unique substituted positions (V109, T345, L369, S372) into NTCP and vice versa; if NTCP retains transport, those substitutions are not causal.
5. **Molecular dynamics** comparing SLC10A4 vs NTCP vs ASBT elevator transitions — tests the dynamic/gating alternative.

Curation Leads (require curator verification)

Action	changes:	-	Keep	NOT	enables	G0:0008508
(						
P 18355966						
<i>Verify snippet:</i> "Slc10a4 showed no transport activity for the Ntcp substrates taurocholate, estrone-3-sulfate, dehydroepiandrosterone sulfate, and pregnenolone sulfate"						
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P 18355966						
Edit any free-text/supporting rationale that attributes orphan status to lost or degenerate Na ⁺ /bile-acid residues. Replace with a statement that the translocation-pathway residues are retained and the cause of orphanhood is unresolved.						

Candidate reference to cite for the structural anchor: -

[P 35545671](#)

— *verify snippet*: "The liver takes up bile salts from blood to generate bile, enabling absorption of lipophilic nutrients and excretion of metabolites and drugs" (source of PDB 7ZYI, the NTCP structure with 2 Na⁺ + cholic acid used to define the coordinating residues).

Suggested curator questions: - Does the current review text imply residue degeneration? If so, flag for correction. - Is any positive transporter MF term asserted anywhere? If so, confirm it is correctly marked **NOT** or removed.

Suggested experiments (for a "future directions" note): cryo-EM of SLC10A4; direct Na⁺/bile-salt binding assays; broad substrate screen; NTCP↔SLC10A4 chimera swaps at the 4 unique positions.

GO decision table:

GO term	Aspect	Current	Recommended action	Basis
GO:0008508 bile acid:sodium symporter activity	MF	NOT enables (IMP, P 18355966)	Retain NOT; remove residue-loss justification	Direct assay negative (P 18355966); residues shown retained (this report)
(any positive symporter/transport MF)	MF	—	Do not add	No positive evidence

Evidence Base (literature)

- **PMID:18355966** — *Cloning and molecular characterization of the orphan carrier protein Slc10a4: expression in cholinergic neurons of the rat central nervous system*. Establishes the no-substrate/orphan phenotype that the seed hypothesis seeks to explain. Verified snippet: "Slc10a4 showed no transport activity for the Ntcp substrates taurocholate, estrone-3-sulfate, dehydroepiandrosterone sulfate, and pregnenolone sulfate." The phenotype is real; the structural analysis shows it is **not** explained by residue loss.

- **PMID:35545671** — *Structural basis of sodium-dependent bile salt uptake into the liver*. Source of PDB 7ZYI (human NTCP with two Na⁺ ions and cholic acid), from which every Na⁺ and bile-acid contact residue in this report was read directly. The structural anchor of the entire analysis. Verified snippet: "The liver takes up bile salts from blood to generate bile, enabling absorption of lipophilic nutrients and excretion of metabolites and drugs."
- **PMID:35580629** — *Structure of the bile acid transporter and HBV receptor NTCP*. Corroborating NTCP structural context (fold, dual function).
- **PMID:42031258** — *Receptor recognition of hepatitis B and D viruses: Structural insight and its application to drug discovery*. Review-level orientation on NTCP mechanism and its bile-acid-transport vs viral-receptor functions.

Proposed Follow-up Actions

1. **Curation edit (immediate):** retain `NOT GO:0008508`; strike any "degenerate/lost residues" rationale and insert the "residues retained, cause unresolved" statement.
 2. **Attach provenance:** the residue-mapping tables (F001/F002), the three-way NTCP/ASBT/SLC10A4 comparison (F004), and the conservation figure as review artifacts.
 3. **Flag for experimentalists:** prioritize a broad SLC10A4 substrate screen and direct binding assays — the structural data say the site is competent, so the field should look for the missing substrate or the broken dynamic/regulatory step, not a broken binding site.
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Prepared from a 3-iteration structure-anchored investigation. Computational mappings are inferences validated by multiple independent methods (direct cryo-EM reading, global sequence alignment, AlphaFold, Phenix superposition) but are not a substitute for an experimental SLC10A4 structure.
